A fusion-junction vaccine in extraskeletal myxoid chondrosarcoma: what can be established today, and the capabilities that would change it

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Author. Tristan D. McRae

Independent researcher, unaffiliated. Correspondence: trimcrae@gmail.com ORCID: 0000-0002-1823-1451

Preprint status. This manuscript is a preprint. It has not been peer reviewed and has not been submitted to a journal. It has not been read by a sarcoma medical oncologist or by a tumour immunologist, and a reader should weigh it accordingly. Independent, personal-capacity work, unconnected to the author's employer; Section 9 states the role of AI tools.

Running title. A junction vaccine in EMC: what is established

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Abstract

Background. Extraskeletal myxoid chondrosarcoma (EMC) is a rare soft-tissue sarcoma defined by rearrangement of *NR4A3*, most often to *EWSR1*. The fusion junction encodes a peptide sequence absent from either parent protein, and because the fusion is the truncal driver it is present in every tumour cell and cannot be lost without loss of the driver. The sponsors of an individualised neoantigen therapy have announced a positive phase 3 result in resected melanoma [9], which makes the platform question timely for other tumours.

Purpose. This paper neither predicts that an EMC vaccine will work nor argues that it will not. It reports what current instruments and access establish about the target, separates limits of the tumour from limits of method or access, and records for each movable limit what would move it.

Methods. Junctions were derived at the transcript level from Ensembl exon structure, so the acceptor exon is retained whole including its 5' untranslated region. Class I binding was predicted with MHCflurry 2.1.4, models release 2.2.0 [2], on a ten-allele panel for the junction screen and a 34-allele panel for the coverage scan, calling a peptide strong at a presentation percentile of 0.5 or below; class II binding with MHCnuggets [12] on three DRB1 alleles at 100 and 1000 nM. Coverage is the union carrier frequency of the presenting alleles over Allele Frequency Net Database records [1]; the sampling model that pooling would require does not hold, so no confidence interval is placed on it and the threshold sensitivity is reported instead. Novelty was assessed by exact-match search against the UniProt reviewed human proteome including isoforms. Clinical figures come from a curated EMC registry. No wet-laboratory data were generated.

Results. Of 27 declared exon pairs, 5 are in frame, yielding 174 junction-spanning peptides and 11 distinct predicted binders of which 4 are strong; there is no pan-EMC epitope. Predicted coverage is a property of the screen as much as of the junction: the commonly reported *EWSR1* exon 7 to *NR4A3* exon 3 junction covers 8.5% on ten alleles, presented on HLA-B*15:01 alone, and 12.3% on 34, where the same lead peptide is also strong on HLA-A*\30:02; pooling every in-frame junction gives 27.4% and 30.4% on those two panels. None is a ceiling — each presenting allele rests on one peptide-allele call, all five of them within 0.1264 percentile units below the acceptance threshold, and a cut anywhere below 0.3736 removes every one. 170 of 174 peptides are absent from the reviewed human

proteome including isoforms and all 4 strong binders survive; the 4 that do not occur in an *NR4A3* isoform, belong to the four aspartate-seam junctions, and cost one predicted binder. The class II arm returns 2 binders and none strong on three DRB1 alleles, only one informative, so combined CD8 and CD4 coverage is not computed rather than zero. The candidate construct is 11 residues carrying class I epitopes only.

What bounds each conclusion. Ten limits are enumerated in Section 3 and graded there as bounded by the disease, by current instruments, or by access. Each movable one is paired with the advance that would move it and the observation that would show it had arrived; no date is offered for any of them.

Interpretation. The most defensible present statement is neither that the route is viable nor that it is closed, but that it is instrument-limited in identifiable ways, and that several of the numbers a reader would take as bounding it are bounding the screen instead. Two findings are offered as results. Seam-proximal peptides of four of the five in-frame junctions reproduce a sequence in a normal *NR4A3* isoform, which withdraws a predicted binder and is a defect in the novelty filter that will recur at any breakpoint whose seam reconstructs an isoform boundary. And the coverage figures this route has been graded on move with the panel and the threshold by more than the distance between them. The paper also observes, of this programme's own route ledger and not of the field, that several priming-directed classes were excluded there for want of antigen supply while a vaccine is an antigen supply, so the combination was never graded here as a unit; the standing objection to the vaccine is not the one that observation answers. Predicted binding is a screen and not evidence of presentation, immunogenicity or benefit, and nothing here supports use of any agent outside a clinical trial.

1. A standing-state report rather than a verdict

A verdict of "unpromising" delivered against a target whose presentation has never been measured records the state of the measuring apparatus, in a form that reads as a statement about the tumour and that nobody revisits when the apparatus improves. This paper separates the three kinds of limit such a verdict conflates. Some are properties of this disease and this junction — a quiet genome, a myxoid matrix that excludes lymphocytes, an incidence below one per million per year — and will not move. Some are properties of today's instruments, chiefly a sequence-based predictor standing in for a measurement nobody has taken. Some are limits of access rather than of knowledge: no published EMC immune profiling, no reachable patient material, no manufacturing route at this incidence. Section 3 records, for each movable limit, the advance that would move it and the evidence that would count as that advance arriving, with no date attached; Section 6 records what would look like each arriving without being it.

The immediate occasion is external. The sponsors of an individualised neoantigen therapy given with pembrolizumab have announced that a randomised phase 3 trial met its primary endpoint of recurrence-free survival in resected stage IIB to IV melanoma [9]; that announcement is a company press release, no effect size was disclosed in it, and the peer-reviewed evidence in this setting remains the phase 2b trial [3]. The result does not transfer to EMC, and Section 4 sets out the axis on which the transfer fails. What it shows is that the manufacturing and delivery apparatus for an individualised RNA vaccine exists as a clinical reality rather than as a proposal.

2. The target and its current evidence base

2.1 Disease and fusion

EMC accounts for roughly 1 to 3% of soft-tissue sarcomas, with an estimated incidence well under one per million per year [7]. It is defined by rearrangement of *NR4A3* on chromosome 9q22. *EWSR1*::*NR4A3* is the commonest fusion, reported in approximately 62 to 75% of cases [7] and in 79% of a molecularly confirmed series of 58 cases

[10]; variant partners include *TAF15*, *TCF12*, *TFG* and *FUS* [7]. The genome is otherwise quiet, and the fusion is the truncal driver.

Two consequences follow, and they point in opposite directions. Because the fusion is truncal and clonal, it is present in every tumour cell and cannot be subclonally lost, so a T-cell response directed at the junction cannot be escaped by antigen loss in the way that a response against a passenger mutation can. Because the genome is quiet, the junction is close to the only tumour-exclusive antigen the disease offers, so if the junction fails there is no second candidate to fall back on. The same feature that makes the target durable makes the portfolio of targets shallow.

2.2 Junction structure and predicted binding

Junctions were derived from the spliced transcripts rather than from an assumed breakpoint, so the acceptor exon is retained whole with its 5' untranslated region, as a fusion transcript retains it. This matters: a superseded model that concatenated coding sequences discarded that retained region and selected a junction disjoint from the one the transcript model produces. All figures below are from the transcript model.

Of 27 declared exon pairs, 5 are in frame: *EWSR1* exons 7, 9, 10, 12 and 13 joined to *NR4A3* exon 3. The remaining 22 are graded out as non-coding acceptor (9), out of frame (4) or not producing the seam (9). The in-frame set yields 174 distinct junction-spanning peptides, screened with MHCflurry 2.1.4, models release 2.2.0 [2], over ten alleles — HLA-A*01:01, A*02:01, A*03:01, A*11:01, A*24:02, B*07:02, B*08:01, B*15:01, B*35:01 and B*44:02 — calling a peptide strong at a presentation percentile of 0.5 or below and weak at 2.0 or below. That screen returns 11 distinct predicted binders, 4 of them strong. The lead candidate at the commonly reported *EWSR1* exon 7 junction is NMPCVQAQY on HLA-B*15:01, at a presentation percentile of 0.37 and a predicted affinity of 73.4 nM. The class call is made on the percentile, not the affinity, and the two do not agree in rank order across this set, so affinities appear below only beside the percentile that classified them.

That ten-allele panel is the instrument behind every binder figure here, and a wider one changes them: a 34-allele screen of the same peptides at the same threshold returns five strong peptide-allele calls rather than four, because NMPCVQAQY is also strong on HLA-A*30:02. Section 2.3 reports both panels.

There is no pan-EMC epitope: the most widely shared candidate appears in 4 of the 5 junctions and is a weak binder, three of the five junctions return no strong binder at all, and every strong binder is specific to its breakpoint. One of the 11 predicted binders, DMPCVQAQY on HLA-B*35:01, is withdrawn on the proteome search reported under B5 below, leaving 10; all 4 strong binders survive that search.

2.3 Population coverage

Coverage here means the union carrier frequency of the presenting alleles, computed as one minus the product of squared non-carrier frequencies over Allele Frequency Net Database records [1]. Two properties of that quantity govern how the figures below should be read, and both are properties of the screen rather than of the tumour.

It moves with the panel. On the ten-allele screen the *EWSR1* exon 7 junction is presented on HLA-B*15:01 alone and covers 8.5%; on 34 alleles the same lead peptide is also strong on HLA-A*30:02 and the junction covers 12.3%. Pooling every in-frame junction gives 27.4% on ten alleles and 30.4% on 34. The four figures are one computation at two panel widths and two junction scopes, not four findings, and the 27.4% and 30.4% pair differs by exactly one allele. Extending the panel further can only raise them, so none of them is a ceiling and this paper does not call any of them one.

It moves with the threshold, further than it moves with anything else. All five strong peptide-allele calls on the 34-allele screen lie between presentation percentiles 0.3736 and 0.4986, against an acceptance threshold of 0.5, and each of the four presenting alleles rests on exactly one of them. Moving the threshold to 0.45 leaves three alleles and

23.2%, to 0.40 leaves one and 8.5%, and to 0.37 leaves none and 0.0%. The whole set spans 0.125 percentile units, sits within 0.1264 of the cut, and a cut below 0.3736 — a move of 0.1264 — takes the headline figure to zero.

What is not reported, and why. Earlier versions printed Wilson 95% intervals about half a percentage point wide on every figure above. They are withdrawn. They were computed by pooling every reference population into one binomial, and the same records show HLA-B*15:01 ranging from 0 to 0.40 in frequency between those populations, so the single-urn model the interval assumes is refuted by its own input — and the interval is an order of magnitude narrower than the threshold sensitivity above. Regional figures are point values for the same reason, and their spread is large: 1.4% in Melanesia and 60% in Northern Europe, on 579 and 923 individuals. In Northern Europe two alleles reach 52.8%, so any statement that no panel reaches half of patients holds for the pooled global frame and not everywhere.

Because an individualised platform selects against the patient's own genotype rather than a public epitope, the relevant figure is the pooled-junction one. Selecting per patient moves it from 8.5% to 30.4% on the panels screened here, and removes neither the panel dependence nor the threshold dependence.

2.4 Limits of the current evidence

Predicted binding is a screen. It is not evidence that any peptide is presented on an EMC tumour cell, not a measure of the density at which it would be presented, and not evidence of T-cell recognition. No EMC immunopeptidomic dataset is known to the author. Sequence-level novelty against the proteome has now been tested and is reported under B5, which excludes one failure mode and leaves presentation, immunogenicity and cross-reactivity untouched. These gaps are enumerated as limits B2 and B3 rather than treated as resolved.

3. The standing state, limit by limit

Ten limits are enumerated, and the second column is the one that matters. Disease-bounded limits are properties of this tumour and are not expected to move; instrument-bounded limits move when methods move; access-bounded limits are properties of this programme's circumstances rather than of anyone's knowledge.

ID	Limit	Bounded by	Best available answer today	What would move it
B1	Predicted class I coverage is low and panel-dependent	instrument, partly disease	8.5% / 12.3% public junction, 27.4% / 30.4% pooled, on 10 / 34 alleles; 0.0% at a 0.37 cut	Wider panels; a defended threshold; measured presentation
B2	Presentation predicted, never measured	instrument and access	MHCflurry 2.0 percentiles on 174 peptides	Immunopeptidomics on EMC tissue or a patient-derived line
B3	Self-adjacency and central tolerance	instrument	Seam structure characterised; no tolerance model applied	Distance-to-self and anchor-versus-contact filters; T-cell reactivity assay
B4	No strong CD4 epitope found on the panel tested	instrument	2 binders, 0 strong, on 3 DRB1 alleles of which 1 was informative; bounds the per-allele rate only at 63%	Wider DR, DP, DQ panel; a class II threshold; measured presentation
B5	Seam-proximal peptides of four junctions occur in an *NR4A3* isoform	instrument, one failure mode resolved	170 of 174 novel proteome-wide; one binder withdrawn	Sequence novelty answered; cross-reactivity is not, and the filter should become isoform-aware
B6	Immunologically cold microenvironment	disease, addressable in combination	Inferred, not measured in EMC	A vaccine supplies antigen; a checkpoint inhibitor supplies release
B7	Physical exclusion by the myxoid matrix	disease	Inferred from histology and pathway expression	Vascular normalisation; matrix-directed agents
B8	No EMC immune profiling published	access	None; the cold and excluded readings are inferences	A deposited EMC series, or a pan-sarcoma atlas reaching this histology
B9	Manufacturing economics at this incidence	access	Five enumerable in-frame junctions, not per-patient discovery	A platform holder; a master-protocol vehicle
B10	Trial design below the randomisation threshold	disease	A 24-patient histology cohort has been run across nine centres	Adaptive or histology-cohort design with a defensible endpoint

3.1 The limit worth watching

The single most consequential of the movable limits is B2. Direct mass-spectrometry evidence that a fusion-junction peptide is presented at measurable abundance on a common allele, in any fusion-driven tumour, would convert the central question of this route from a prediction into an empirical one. The coverage instrument used here is disclosed as failing for exactly that reason: it computes an eligibility fraction for an epitope whose presentation is unestablished. No date is offered for that arrival, or for any other in this section. An earlier version of this paper carried a table of optimistic, expected and conservative years for four such capabilities; it is withdrawn, because a forecast a reader cannot check does not become checkable by being tabulated, and Section 6 does the work it was there for by saying what each arrival would and would not look like.

B1. Low predicted class I coverage, dependent on the screen

Proposition. On the panels and thresholds screened here, fewer than a third of patients carry an allele predicted to present any junction peptide — and that fraction is set by the screen at least as much as by the tumour.

Evidence. The 34-allele screen finds 4 presenting alleles and 30.4%; the ten-allele screen finds 3 and 27.4%. Both rest on five strong peptide-allele calls, one per presenting allele, and Section 2.3 gives their threshold sensitivity.

What would clear or move it. Three things, in ascending cost. Extending the panels beyond 10 and 34 class I alleles to the full validated set is a computational task and would raise the denominator. The acceptance threshold is a convention rather than a result — nothing here defends 0.5 against 0.4 or 0.6 — and settling what the cut should be for junction peptides would do more to this figure than any panel extension. Measured immunopeptidomics (B2) could promote peptides the predictor ranks weakly or remove ones it ranks strongly, in a direction not knowable in advance.

What would not move it. Manufacturing improvements, delivery formulation and adjuvant selection change nothing here. This limit is about which patients have a target at all.

Residual. Some fraction of patients will have no presented junction peptide and no second antigen to substitute. That fraction is a real and permanent exclusion from this approach, and it should be stated in any protocol rather than discovered at screening.

B2. Predicted rather than measured presentation

Proposition. No junction peptide has been shown to be presented on the surface of an EMC cell.

Evidence. All binding figures in Section 2 come from MHCflurry 2.0, a sequence-based predictor [2]. No EMC immunopeptidomic dataset is known to the author, and the repository contains none.

What would clear it. Mass-spectrometry immunopeptidomics on EMC tumour tissue or on one of the patient-derived EMC cell lines that have been established and characterised [6,14]. A single positive identification of a junction-spanning peptide in an EMC eluate would convert the central premise of this route from predicted to observed. A negative result across adequate material would be close to decisive against the route, which is what makes this the highest-value single experiment in the ledger.

Cost and owner. Requires tissue and a proteomics facility. It is not computational and cannot be performed by this programme.

B3. Self-adjacency and central tolerance

Proposition. The junction peptide is mostly self sequence with one or two novel residues at the seam, so the T-cell repertoire capable of recognising it may have been deleted or anergised.

Evidence. At the *EWSR1* exon 7 junction the seam carries a single novel codon formed from one leftover *EWSR1* nucleotide and two retained acceptor 5' untranslated nucleotides, after which *NR4A3* methionine 1 follows as an internal residue. The remainder of each peptide is parental sequence. Whether one altered residue is sufficient to break tolerance is not answerable by binding prediction.

What would clear or narrow it. Two computational filters were specified for this route and never built: a distance-to-self and tolerance filter, and an analysis of whether the novel residues fall at anchor positions, which affect binding, or at positions contacting the T-cell receptor, which affect recognition. A peptide whose only novelty is at an anchor may bind differently from self without looking different to a T cell, which is the failure mode that matters here. Both filters are computational and neither has been run. Beyond that, only a T-cell reactivity assay against the specific peptide-HLA complex answers the question.

Comparator. This limit is where EMC differs most sharply from the melanoma setting, in which selected neoantigens frequently arise from point mutations in a repertoire that has not been tolerised against them and in a tumour already under immune pressure.

B4. No strong CD4 epitope on the three alleles tested

Proposition. An effective vaccine generally requires CD4 helper epitopes, and the corrected junction supplies no strong class II binder on the three alleles screened. That is a statement about the screen, and it is reported here as one.

History. This arm was previously withheld because it sat on the superseded coordinate system, on a seam disjoint from the class I set. The builder has since been moved to the transcript model and the arm regenerated, so the two arms are now certified to sit on the same seam.

Result. With the class II and class I arms now certified to sit on the same seam, the arm can be reported. At the corrected *EWSR1* exon 7 junction, 15 candidate 15-mers were screened with MHCnuggets [12] against DRB1*15:01, DRB1*\03:01 and DRB1*07:01, calling a peptide a binder below 1000 nM and strong below 100 nM. Two peptides bind and none is strong: YSQQSSSYGQQNMPC at 262 nM and QYSQQSSSYGQQNMP at 439 nM, both on DRB1*\07:01. The superseded version on the retracted seam reported 9 binders of which 4 were strong. The correction weakened this arm as it weakened the class I arm.

What that result does and does not bound. Very little, for four reasons. The 15 candidates are single-residue-offset windows over one seam sharing 14 of 15 residues, so the number of independent peptides tested is nearer one than fifteen. Two of the three alleles produced nothing within an order of magnitude of a threshold, at 3,061 nM and 13,585 nM, so the panel is effectively one allele wide. Zero of three bounds the per-allele probability of a strong binder at 63% from above and no lower. And the 100 nM cut is a class I convention, where the conventional class II binder band is 500 to 1000 nM, inside which 262 nM sits comfortably. The negative describes what this screen returned; it is not evidence that the junction supplies no helper epitope.

What would clear or move it. The panel is three alleles, which is narrow, and class II presentation is substantially harder to predict than class I; a wider DR, DP and DQ panel is a computational task and may change the picture. Beyond that, only measured class II presentation settles it. A construct can also supply help from a heterologous source rather than from the junction itself, which is standard practice in peptide vaccine design and would make this limit a design constraint rather than a blocking one.

Consequence for the construct, and for the combined figure. Two things follow directly. The candidate construct regenerated at the corrected junction carries two strong class I epitopes, NMPCVQAQY and QQNMPCVQAQY, both on HLA-B*15:01, and no class II epitope at all; its minimal synthetic long peptide is 11 residues where the retracted-seam version was 27 residues carrying both arms. And the combined CD8 and CD4 coverage figure is not computed, which is different from being zero. The coverage instrument defines class II coverage over the alleles presenting a strong class II binder; with none, its class II branch never evaluates and the field it writes is empty. That empty field records that the quantity was not computed, and the instrument's own note describes the class II coverage it would compute as a floor over a tested panel that untested alleles could only raise. Reporting it as a null result would be reading an absent value as a measured zero.

Status. Reported rather than withheld, negative on the three alleles tested, and bounding the general availability of helper epitopes at this junction hardly at all.

B5. Four peptides occur in an *NR4A3* isoform

Proposition, as originally stated. The junction peptides had been tested for novelty against two proteins rather than against the proteome, so their absence from normal human proteins was not established.

Result. All 174 distinct junction peptides were searched by exact substring against the UniProt reviewed human proteome, isoform sequences included. 170 are absent from every reviewed human protein. All 4 strong binders survive, including the *EWSR1* exon 7 lead NMPCVQAQY. The limit is largely cleared, and the sequence-level novelty premise of this route holds for the great majority of the peptide set.

The four that do not, and which junctions they belong to. DMPCVQAQ, DMPCVQAQY, DMPCVQAQYS and DMPCVQAQYSP all occur in Q92570-3, an isoform of *NR4A3* itself. One of them, DMPCVQAQY, is a predicted binder on HLA-B*35:01 at 369.1 nM. Those four peptides are not tumour-exclusive, and DMPCVQAQY is withdrawn as a candidate.

The pattern is not random across the junction set. All four belong to the same four junctions, *EWSR1* exons 9, 10, 12 and 13, which are the four whose seam codon is aspartate; the *EWSR1* exon 7 junction has an asparagine seam codon and none of its peptides collides. So four of the five in-frame junctions share a seam whose most seam-proximal peptides reproduce a normal *NR4A3* isoform sequence, and the one junction that is clean is the commonly reported public one that carries the lead binder. Within the affected junctions the collision is confined to peptides that begin at the seam residue, which are the peptides with the least donor content; those extending further into *EWSR1*, including the strong binder RGDMPCVQAQY, remain novel. The consequence for design is specific rather than general: for the aspartate-seam junctions, a construct should not rely on the seam-proximal window.

The methodological finding. The upstream novelty filter compares each candidate against the canonical parent proteins only, so an isoform that carries the seam sequence passes it unseen. That is a defect in the filter rather than in these particular junctions, and it will recur for any breakpoint whose seam residue reconstructs an isoform boundary. The proteome search reports this condition explicitly rather than silently discarding the hits. The filter should be made isoform-aware.

What a clean result does not license. A peptide absent from every reviewed human protein is not thereby safe. A T-cell receptor engages a peptide-MHC surface rather than a sequence, so a peptide differing from a self peptide at a position that does not contact the receptor can still be cross-recognised. Unreviewed sequences were not searched: a hit among predicted-and-unreviewed entries is not evidence that a normal protein carries the peptide, and a miss there is not evidence of absence. This test excludes one specific failure mode and leaves the others standing.

B6. Immunologically cold microenvironment

Proposition. EMC has a low mutational burden and a sparse infiltrate, so there is little pre-existing antigen-specific response for an intervention to amplify.

Evidence, and an observation about this programme's own grading. This proposition is why this programme set most immune-modulating classes aside for this disease in its route ledger, a committed record of forty candidate modalities each carrying a verdict and a reason. Those reasons are quoted below from that ledger: they are this author's own prior notes, not positions taken in the literature. Innate agonists of the STING, TLR and RIG-I classes were excluded there because such agonism "supplies the danger signal and the priming context; it does not supply antigens, and a genome as quiet as EMC's is short of antigens rather than short of priming". In-situ vaccination was excluded because it "releases and adjuvants the antigens the tumour already has", so "releasing more of nothing does not help". Checkpoint inhibitors beyond PD-1, costimulatory agonists, adenosine-axis inhibitors and regulatory T-cell depletion were each excluded for acting on a response presumed absent.

Every one of those is an argument about antigen supply, and a vaccine is an antigen supply, so none of them argues against the class in a configuration where antigen is supplied exogenously. That is why the combination in Section 4 was never graded here as a unit.

The symmetry is not exact, and an earlier version of this paper asserted that it was. The vaccine was not parked for want of priming: its ledger entry is on the board rather than excluded, and its stated reason for being parked is immunogenicity, "which its own record states is not a computational question". The standing blocker against it names two things, that EMC is antigen-cold and that the fusion junction is a weak peptide-HLA, and the second is an antigen-side objection no priming-directed partner answers. The correct statement is the weaker one: a partner supplies what several exclusions called missing, and does not supply what the vaccine's own blocker calls missing.

What would clear it. Clinical evaluation of a vaccine together with an agent that supplies priming or releases inhibition. Section 4 sets out the specific backbone for which EMC evidence already exists.

B7. Physical exclusion by the myxoid matrix

Proposition. EMC is immune-excluded by a dense chondroitin-sulfate gel, which is a physical barrier rather than a signalling programme.

Evidence. The myxoid matrix is the disease's defining histological compartment. The oncofetal chondroitin sulfate pathway is the mechanism proposed for it here by analogy: the glycosaminoglycan biosynthesis genes of that pathway are differentially expressed and correlated with immune response in placenta and colorectal cancer [8], which is the tissue setting that study examined. No EMC-specific expression evidence for the pathway is cited, because none is known to the author, and the inference from those tissues to this one is the author's. Transforming growth factor beta inhibition, which is the standard proposal for immune-excluded tumours, was set aside for EMC precisely on the ground that the exclusion here is physical rather than driven by a fibroblast programme.

Why this limit is distinct from B6. A cold tumour lacks a response. An excluded tumour may have a response that cannot reach the target. These require different interventions, and an intervention aimed at one does not address the other. A vaccine can raise the circulating frequency of junction-specific T cells without changing whether those cells can enter the tumour.

What would clear or mitigate it. Vascular normalisation is the mechanism with the most direct EMC evidence, discussed in Section 4. Matrix-directed approaches, including addressing the oncofetal chondroitin sulfate modification itself, are registered in this programme as candidates and are not resolved.

B8. Absent EMC immune-profiling data

Proposition. The characterisations of EMC as cold and as excluded are inferred from the disease's mutational burden, its histology and sarcoma-wide immunotherapy experience, rather than from published EMC-specific immune profiling.

Consequence. Several exclusions above rest on an assumption that has not been measured in this disease. The absence of a reading is not a reading of absence, and it is possible that EMC is less cold than assumed, or excluded in a manner that suggests a specific intervention.

What would clear it. Infiltrate quantification, HLA class I expression status and antigen-presentation machinery assessment on a series of EMC specimens. HLA class I loss would independently disable every antigen-directed route including this one, and is not known for this disease.

Cost and owner. Requires tissue. This is the cheapest tissue-based item in the ledger and the one that most efficiently informs the others.

B9. Manufacturing economics at this incidence

Proposition. An individualised vaccine requires per-patient sequencing, design and manufacture, and EMC's incidence is well under one per million per year.

Consequence. No independent programme can supply the manufacturing, and the economics that support an individualised product in melanoma do not obviously extend to a disease with this incidence.

What would mitigate it. Two features of this target reduce the requirement relative to a general individualised product. First, the antigen is not discovered per patient by tumour sequencing but is determined by which exon pair the patient carries, of which five are in frame; the design space is therefore small, enumerable in advance, and shared across patients with the same breakpoint. Second, because the fusion is truncal, the antigen does not need re-selection over the disease course. A small fixed panel of breakpoint-specific constructs, allocated by a diagnostic assay, is closer to a stratified product than to a bespoke one. Whether that is commercially tractable is a question for a platform holder and not one this analysis can answer.

B10. Trial design below the randomisation threshold

Proposition. At this incidence a conventional randomised trial of a vaccine addition is not feasible.

Evidence and mitigation. A histology-specific EMC cohort within a sarcoma master protocol has already been executed and reported for a different regimen, enrolling 24 patients across nine centres in three countries over four years [5]. That shows the vehicle exists, and it gives an accrual rate of about six unselected patients a year across nine centres.

And that rate does not survive this paper's own eligibility filter. A junction-vaccine cohort cannot enrol unselected EMC: it needs *EWSR1*::*NR4A3*, which is 62 to 75% of cases [7], and then an HLA type that presents a junction peptide, which is 8.5 to 30.4% of those (B1). Applying both fractions to that trial's own accrual leaves roughly 0.3 to 1.4 eligible patients a year, so a cohort of the size already run would take on the order of two decades, and a powered comparison considerably longer. That is what makes B10 a limit of the disease rather than of study design: the vehicle exists and the patients to put in it do not arrive fast enough. The endpoint question is separate — response-based endpoints behave poorly in indolent tumours, and the existing EMC cohort reported progression-free survival at a fixed time point.

4. An ungraded combination

The limits above do not resolve independently. B6 and B7 are properties of the tumour that a vaccine cannot address, and B1 and B3 are properties of the antigen that a checkpoint inhibitor cannot address. The relevant question is therefore not whether a junction vaccine works in EMC, which is the question that was asked and answered negatively, but whether a junction vaccine adds anything to a backbone that already has EMC-specific activity.

Such a backbone exists. A phase 2 histology-specific EMC cohort within the IMMUNOSARC II master protocol evaluated sunitinib with nivolumab in adults with advanced, progressing, centrally confirmed EMC across nine centres in Spain, Italy and the United Kingdom. Of 23 evaluable patients, 16 were progression-free at 6 months, median progression-free survival was 13.2 months (95% CI 5.7 to 20.7), and there were 2 partial responses [5]. This is a conference abstract and has not been peer-reviewed; the full publication is not available, and no results are posted for the registration. It is reported here as the most direct EMC-specific evidence available for an immunotherapy-containing regimen, and not as evidence of efficacy. The preceding mixed-histology phase Ib/II study of the same combination is published [4].

Three features make this backbone the natural context for the vaccine question. The antiangiogenic component addresses B7 by a mechanism — vascular normalisation reduces the physical and vascular barriers to lymphocyte entry — and antiangiogenic tyrosine kinase inhibitors carry the most consistent prospective signal in this disease: pazopanib gave an objective response in 4 of 22 evaluable patients with a median progression-free survival of about 19 months [11], and a sunitinib series reported activity in translocated EMC [13]. The checkpoint component addresses the release arm of B6. Neither addresses antigen supply, which is what the vaccine would contribute.

This is also the architecture of the melanoma programme behind the recent announcement, where the individualised vaccine is given with pembrolizumab and never alone [3,9]. The transfer to EMC fails on antigen depth — melanoma supplies a pool of private neoantigens, EMC one junction — and not on architecture.

What this section is claiming, and what it is not. The proposition is that adding a breakpoint-matched junction construct to a checkpoint and antiangiogenic backbone gives a combination in which each component covers a limit the others do not, and that it has been graded as a unit nowhere — including in this programme's own ledger, which is where B6's observation comes from. It is not a trial proposal. No population, comparator, endpoint, effect size or sample size is specified here; a single-arm addition to a backbone whose own six-month progression-free rate is 16 of 23 could not attribute an effect to the vaccine; and B10's arithmetic says the eligible patients do not arrive at a workable rate. Whether the combination merits evaluation at all turns on B2. Nothing here recommends that any patient receive any of these agents outside a clinical trial.

5. Present work and pending arrivals

Four items on this list need nothing but public data and would sharpen numbers this paper reports as provisional: the distance-to-self and anchor-versus-contact-position filters specified for this route and never implemented, which bear directly on B3; extension of the 10- and 34-allele class I panels and the three-allele DR panel to the full set for which validated predictors exist; a defended acceptance threshold for junction peptides, without which B1's figures remain as sensitive to the cut as Section 2.3 shows them to be; and the isoform-aware novelty filter identified in B5. None requires permission, material or funding.

Everything else waits. Measured presentation (B2) would change the character of this route rather than its score, because the screen in Section 2 would stop being a stand-in and become a hypothesis with a calibrating dataset. A deposited EMC expression or proteomics series would settle B8, which is the cheapest item that informs the most others, since infiltrate density and HLA class I expression status bear on every antigen-directed route in this disease. Access to a patient-derived line would additionally make B2 executable rather than merely specified; these are separate arrivals and neither implies the other. B6 and B7 are properties of the tumour that no computational advance addresses at all.

6. Distinguishing a real advance from an apparent one

Each capability this route waits on has a plausible near-miss that the literature would report in language resembling a hit.

A result on a different fusion is not a result on this one. A presentation or immunogenicity study on another fusion-driven tumour raises the prior that fusion junctions in general can be seen by T cells. It does not establish that this junction, on these alleles, is presented at usable abundance. The discriminating question to ask of any such report is whether its evidence concerns presentation, abundance, or only a different fusion.

Robotic execution is not material access. A cloud laboratory with per-experiment pricing supplies robots and generic reagents, not an EMC line or organoid, and without that none of the tissue-gated items in Section 3 becomes runnable.

A better predictor is not a measurement. An improved class I or class II model would change the numbers in Section 2 and leave B2 where it is. The limit there is that nobody has looked, not that the looking-glass is imprecise.

A larger allele panel raises a figure without grounding it. Extending the panels can only move coverage upward, and a higher predicted-presentation figure is not evidence that any patient's tumour presents anything.

6.1 Conditions for revision

Four statements in this paper are falsifiable by an observation a reader could go and make, and they are listed so a future reader can check them rather than re-derive them. Sequence-level novelty for 170 of 174 peptides would be overturned by a proteome release adding a protein that contains one of the four strong binders. Every coverage figure would move upward if a wider panel added a presenting allele and downward if measured presentation removed one of the strong calls, and all of them go to zero if the acceptance threshold is set below 0.37. The class II statement would be changed by a wider DR, DP and DQ panel, which is the single most likely source of a change to it. And the argument in Section 4 would be closed altogether by an EMC immune-profiling series showing HLA class I loss, which would end every antigen-directed route in this disease including this one.

7. Limitations

All binding figures are predictions from sequence-based models and no EMC-specific validation of either model exists. Coverage is computed by multiplying non-carrier frequencies across alleles, which assumes independence not only between loci but between alleles at the same locus; two alleles pooled here are both HLA-B, and handling that correctly moves the pooled figure by about 0.3 percentage points. The population-to-region mapping is an approximation, regional figures are point values on samples as small as 579 individuals, and the frequencies are pooled across populations whose allele frequencies differ by more than the figures being reported. The binder counts and every coverage figure derived from them depend on an acceptance threshold this paper does not defend, and no multiplicity correction is applied anywhere: 174 peptides were screened against 10 and then 34 alleles at two thresholds, with no decoy control and no null expectation, so the calls that pass are reported as what the screen returned rather than as an enrichment over chance.

The IMMUNOSARC II EMC cohort result is a conference abstract, single-arm and not peer reviewed, and it evaluates a combination whose component with the larger independent EMC evidence base is the tyrosine kinase inhibitor rather than the checkpoint inhibitor; it is cited to establish that a vehicle and a backbone exist, not to attribute activity to the immune arm. The characterisations of EMC as cold and as immune-excluded rest on inference rather than on published EMC-specific immune profiling, which is limit B8. The class II panel is three DR alleles with no DP or DQ, so its negative bounds a narrow question rather than the general availability of helper epitopes. No claim is made that any peptide is presented, that any construct would be immunogenic, that any combination would be safe or effective, or that any of this is ready for clinical use. No wet-laboratory work was performed, and the measurements this characterisation most needs require work this programme cannot carry out.

8. Reproducibility

Every figure in Sections 2 and 3 is generated by a script in `research/modalities/` and is committed as a JSON artifact beside it: `fusion_breakpoints.py` for the junction set and predicted binders, `hla_coverage.py` for population coverage, `coverage_scan.py` for the broad-panel screen and curve, `junction_proteome_novelty.py` for the proteome search of Section B5, `patient_neoepitopes.py` and `patient_cd4_epitopes.py` for the per-patient shortlisters, and `vaccine_construct.py` for the candidate construct and its minimal synthetic long peptide. The corresponding artifacts are `fusion-breakpoint-neoantigens.json`, `hla-coverage.json`, `coverage-curve.json`, `epitope-allele-matrix.json`, `junction-proteome-novelty.json`, `patient-cd4-demo.json` and `vaccine-construct.json`. The predictor versions those artifacts record are MHCflurry 2.1.4 with models release 2.2.0 for class I and MHCnuggets for class II.

These are not regenerated on every commit, and this paper does not claim they are. The workflow that runs them is dispatched by hand, its steps do not fail the run when a generator fails, and it writes its outputs to a separate cache branch from which they are copied in. So the artifacts in the repository are the record of a run that happened, verifiable by their embedded timestamps and input hashes, rather than the output of continuous re-execution. An earlier version of this section said they were regenerated in continuous integration; that was not true of any branch a reader would fetch.

Clinical figures are quoted from the curated EMC registry at `research/data/emc-clinical-registry.json`, which carries a structured citation entry for every source and a retrieval note for those whose identification required one.

9. Declarations

Use of AI tools. A large language model (Claude, Anthropic) was used throughout this work: to write the analysis code, to run the screening pipelines, to draft and revise this manuscript, and to conduct the internal adversarial review of earlier drafts whose findings this version incorporates. The model versions used over the span of the work are the ones the repository's commit record names, and are not restated here. No quantitative result was generated by a language model directly: every figure in Sections 2 and 3 is produced by the code named in Section 8 and is reproducible from the committed artifacts, and the clinical figures are transcribed from the publications cited for them. Every literature identifier in Section 10 was checked against a retrieved bibliographic record held in this repository, and any identifier that could not be so anchored was removed rather than retained. The author takes full responsibility for all content, including for the correctness of the code and for the interpretation of the results.

Data and code availability. All code and all artifacts underlying every figure are in the public repository that accompanies this manuscript, under the paths given in Section 8. No restricted data were used.

Competing interests. The author declares no financial competing interests: he holds no position, equity, consultancy or patent relating to any gene, sequence, peptide or agent named here. One non-financial interest is declared: the author is a survivor of extraskeletal myxoid chondrosarcoma, the disease this work addresses.

Funding. This work received no external funding and was self-funded by the author. No funder had any role in the analyses, the interpretation of the results, or the decision to publish.

Ethics. No human subjects, human material or animals were involved. No patient data were used; the clinical figures quoted are published or publicly presented aggregate results.

Not clinical guidance. Nothing in this manuscript is medical advice, and nothing in it is evidence that any agent or combination is safe or effective in extraskeletal myxoid chondrosarcoma. No peptide reported here has been synthesised, formulated, or tested in any cell, tissue or animal by anyone, and none may be administered to any person. The agents named in Section 4 are discussed as a research hypothesis and their use outside a clinical trial is not supported by anything in this paper.

10. References

Every identifier below is transcribed from a bibliographic record — either one held in this repository's literature and registry files, or one retrieved from Europe PMC or Crossref by the verification workflow that accompanies this work. None is written from recollection.

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2. doi:10.1093/nar/gkz1029. Accessed via the MIT-licensed `slowkow/allelefrequencies` mirror.
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4. Weber JS, Carlino MS, Khattak A, Meniawy T, Ansstas G, Taylor MH, et al. Individualised neoantigen therapy mRNA-4157 (V940) plus pembrolizumab versus pembrolizumab monotherapy in resected melanoma (KEYNOTE-942): a randomised, phase 2b study. *Lancet* 2024;403(10427):632-644. doi:10.1016/S0140-6736(23)02268-7. PMID 38246194.
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Frontiers in Cell and Developmental Biology 2021;9:763875. doi:10.3389/fcell.2021.763875. PMID 34966741. The tissues studied are placenta and colorectal cancer, not sarcoma.

10. Merck and Moderna announce that the phase 3 INTerpath-001 trial of intismeran autogene plus pembrolizumab met its endpoints of recurrence-free survival and distant metastasis-free survival in completely resected stage IIB-IV melanoma. Company press release, 2026. **This is an announcement, not a publication**: no effect size was disclosed in it, none is quoted here, it carries no digital object identifier, and it is not indexed in any bibliographic database this work can query. It is cited only for the fact that the announcement was made.
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12. Stacchiotti S, Ferrari S, Redondo A, Hindi N, Palmerini E, Vaz Salgado MA, et al. Pazopanib for treatment of advanced extraskeletal myxoid chondrosarcoma: a multicentre, single-arm, phase 2 trial. *The Lancet Oncology* 2019. doi:10.1016/S1470-2045(19)30319-5. PMID 31331701. Registration NCT02066285; 26 patients started treatment and 22 were evaluable for the primary endpoint.
13. Shao XM, Bhattacharya R, Huang J, Sivakumar IKA, Tokheim C, Zheng L, et al. High-throughput prediction of MHC class I and II neoantigens with MHCnuggets. *Cancer Immunology Research* 2020. doi:10.1158/2326-6066.CIR-19-0464. PMID 31871119. This is the class II predictor used for the CD4 arm of Section B4. The artifact it produced records neither a tool version nor a models release, where the class I artifact records both; that gap is stated in Section 8 rather than papered over.
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Data sources cited as resources rather than as publications. Transcript structures are Ensembl records for *EWSR1* and *NR4A3*, retrieved and cached as committed inputs. The proteome searched in Section B5 is UniProtKB reference proteome UP000005640, reviewed entries with isoforms included, 42,547 sequences and 24,513,032 residues at retrieval, fetched from the UniProt REST stream; the isoform that carries the four colliding peptides is Q92570-3.

Appendix A. Superseded figures

The following values were reported before the 2026-08-07 coordinate-system correction and must not be quoted. They are retained so that earlier drafts and derived documents can be identified.

Quantity	Superseded value	Current value
In-frame junctions	7	5
Distinct predicted binders	26	11
EWSR1 exon 7 junction, presenting alleles	A*11:01 and B*\08:01	B*15:01 on 10 alleles; B*\15:01 and A*30:02 on 34
EWSR1 exon 7 junction coverage	29.7%	8.5% on 10 alleles, 12.3% on 34
Any-strong-binder coverage	58.0%	27.4% on 10 alleles, 30.4% on 34
Regional range	36% to 79%	1.4% to 60%
Broad-panel presenting alleles	20 of 34	4 of 34
Broad-panel coverage ceiling	84.5%	30.4%, and not a ceiling
Combined CD8 and CD4 coverage	16.5%	not computed
Candidate minimal synthetic long peptide	27 residues, both arms	11 residues, class I only
Class II predicted binders at the *EWSR1* exon 7 junction	9, of which 4 strong	2, of which 0 strong

The superseded values arose from a model that concatenated coding sequences and thereby discarded the acceptor exon's retained 5' untranslated region. The corrected junction set is disjoint from the superseded one, so the earlier peptide identifiers do not appear in the current artifacts.

Appendix B. Statements withdrawn by the first adversarial review of this manuscript

This manuscript was reviewed on 2026-08-22 by five independent blind readers against a pinned commit, each with a different lens, none able to see the others. The statements below stood in the version they read and do not stand now. They are recorded because a reader who met an earlier copy, or a derived document that quoted one, needs to be able to identify what changed and why — and because a correction that leaves no trace is indistinguishable from a claim that was never made.

Withdrawn statement	Why it does not stand
"presented on HLA-B*15:01 alone", stated without naming a panel	True of the ten-allele screen only. The 34-allele screen finds the same lead peptide strong on HLA-A*30:02, and the junction's coverage is then 12.3% rather than 8.5%.
The 30.4% figure described as a "ceiling"	Sections 5 and 6 of the same manuscript said extending the panel could only raise it. It is a union over four alleles at one threshold, and moving that threshold to 0.37 takes it to zero.
"It does not attain 50% at any panel size" — withdrawn	No search over panel sizes was performed; the panel is fixed at 34 and the scanned variable is the number of presenting alleles. In Northern Europe two alleles reach 52.8%.
Wilson 95% confidence intervals on every coverage figure	They were computed by pooling every reference population into one binomial. The same records show the frequency of one pooled allele ranging from 0 to 0.40 between those populations, so the model the interval assumes is refuted by its own input.
"the combined CD8 and CD4 figure is null rather than unreported. That is a result" — withdrawn	The instrument's class II branch never evaluates when no allele qualifies, so the empty field records "not computed". The instrument's own note calls the class II coverage it would compute a floor that untested alleles could only raise.
"The one substantive reasoning correction this work offers …", and the symmetry it asserted	The quoted exclusions are this author's own route-ledger notes, not positions in the literature, and the vaccine's own entry in that ledger is parked on immunogenicity rather than on absent priming. The claim is narrowed in B6 and the Abstract accordingly.
The dated capability bands of the former Section 3.1 (2027H2 / 2029 / 2032)	Withdrawn entire. They were self-described as the weakest material in the paper, and a forecast a reader cannot check does not become checkable by being tabulated.
"regenerated in continuous integration", of the artifacts in Section 8	The workflow is dispatched by hand, its steps do not fail the run when a generator fails, and it writes to a cache branch. Section 8 now says what is actually true.
Reference 8 quoted as "correlated with disease outcome"	The study's title names immune response in placenta and colorectal cancer, which is the tissue setting it examined.
References 10 and 11 as "[citation to verify]"	Both were resolvable in this repository at no cost and are now written out. The one reference that genuinely has no record — the class II predictor — says so in its own entry instead.
"24 patients across nine centres … gives a realistic accrual rate"	Unfiltered by this paper's own eligibility criteria. Applying them gives roughly 0.3 to 1.4 eligible patients a year.